

Информация

Докладчик

- Оффей Эндрю
- Студент физмат
- Российский университет дружбы народов
- Andyoffei@yahoo.co.uk

Цель работы

Целью данной лабораторной работы является ознакомление с основами включения графики в документы LaTeX.

The purpose of this lab work is to learn how to include and manipulate graphics in LaTeX documents using the `graphicx` package and related tools.

Задание

1. Study basic image inclusion with `graphicx` package
2. Learn to modify graphic appearance (size, rotation, scaling)
3. Understand float environments for image placement
4. Practice file naming and organization best practices
5. Learn cross-referencing for figures
6. Explore different float types and positioning options
7. Complete the exercises with practical examples

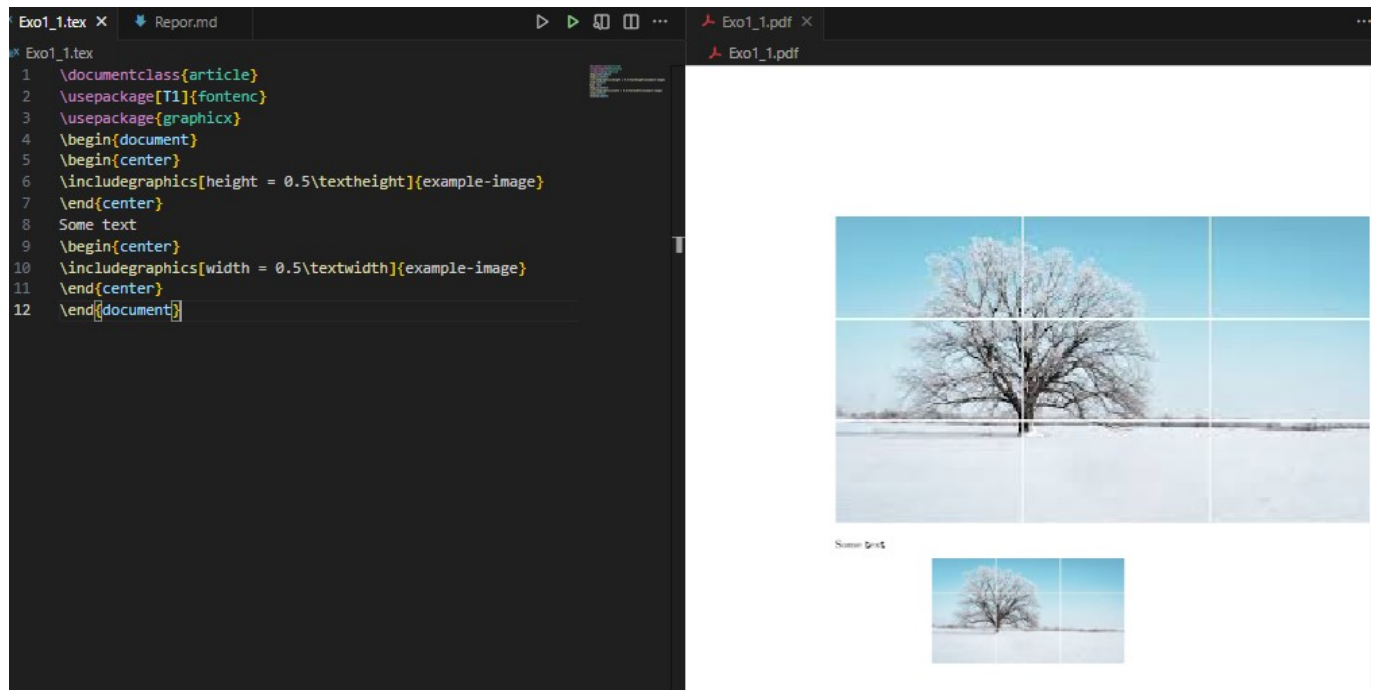
Теоретическое введение

4 Включение графики / Including Graphics

Для включения внешних изображений в LaTeX используется пакет `graphicx`, который предоставляет команду `\includegraphics`. To include external images in LaTeX, use the `graphicx` package which provides the `\includegraphics` command.

4.1 Изменение внешнего вида графики / Altering Graphic Appearance

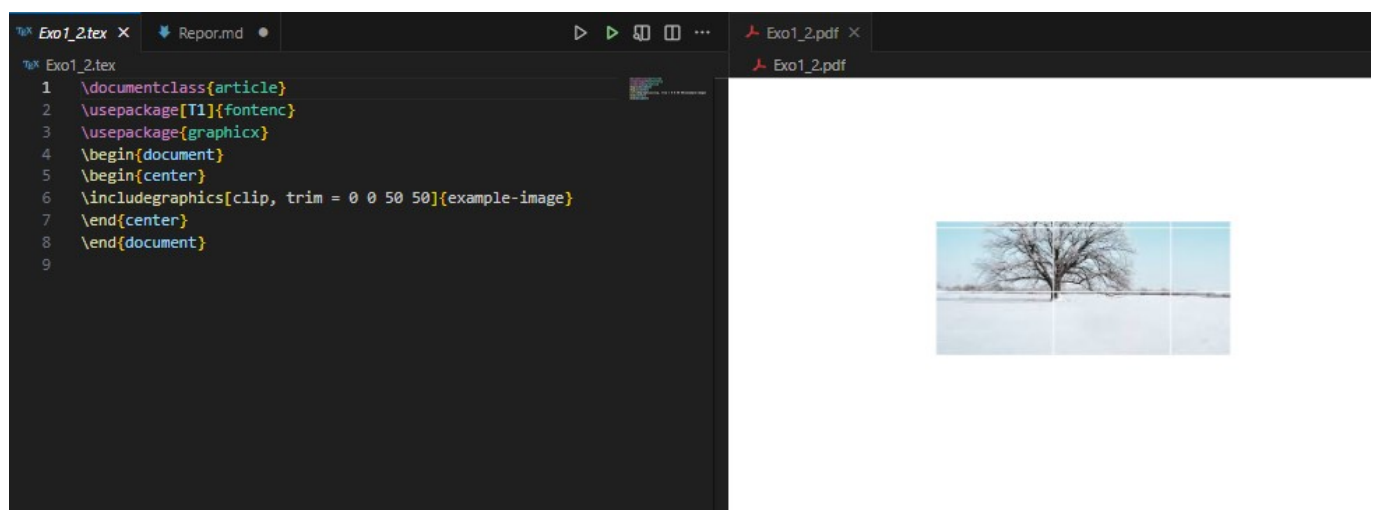
Команда `\includegraphics` имеет множество опций для управления размером и формой изображений. The `\includegraphics` command has many options to control image size and appearance.



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4.2 Создание плавающих изображений / Making Images Float

Изображения обычно включаются как плавающие объекты (floats) чтобы избежать больших пробелов на странице. Images are typically included as floats to avoid large gaps on the page.



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4.3 Размещение плавающих объектов / Placing Floats

Пакет `float` предоставляет опцию `H` для точного размещения плавающих объектов. The `float` package provides the `H` option for precise float placement.

Exo2.tex > {} Figure: An example image

1 \documentclass{article}

2 \usepackage[T1]{fontenc}

3 \usepackage{graphicx}

4 \usepackage{lipsum} % produce dummy text

5 \begin{document}

6 \lipsum[1-4] % Just a few filler paragraphs

7 Test location.

8 \begin{figure}[ht]

9 \centering

10 \includegraphics[width=0.5\textwidth]{example-image}

11 \caption{An example image}

12 \end{figure}

13 \lipsum[6-10] % Just a few filler paragraphs

14 \end{document}

15

Build with agent mode

AI responses may be inaccurate.

Generate Agent Instructions to onboard AI onto your codebase.

[1-4] Test location. [6-10]



Figure 1: An example image

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4.4 Перекрёстные ссылки / Cross-referencing

Механизм `\label` и `\ref` позволяет создавать ссылки на пронумерованные элементы. The `\label` and `\ref` mechanism allows creating references to numbered elements.

Exo6.tex > ...

1 \documentclass{article}

2 \usepackage[T1]{fontenc}

3 \usepackage{graphicx}

4 \usepackage{lipsum} % dummy text for filler

5 \usepackage{float}

6 \begin{document}

7 \lipsum[1-7]

8 \begin{figure}[H]

9 \centering

10 \includegraphics[width=0.5\textwidth]{example-image}

11 \caption{An example image}

12 \end{figure}

13 \lipsum[8-15]

14 \end{document}

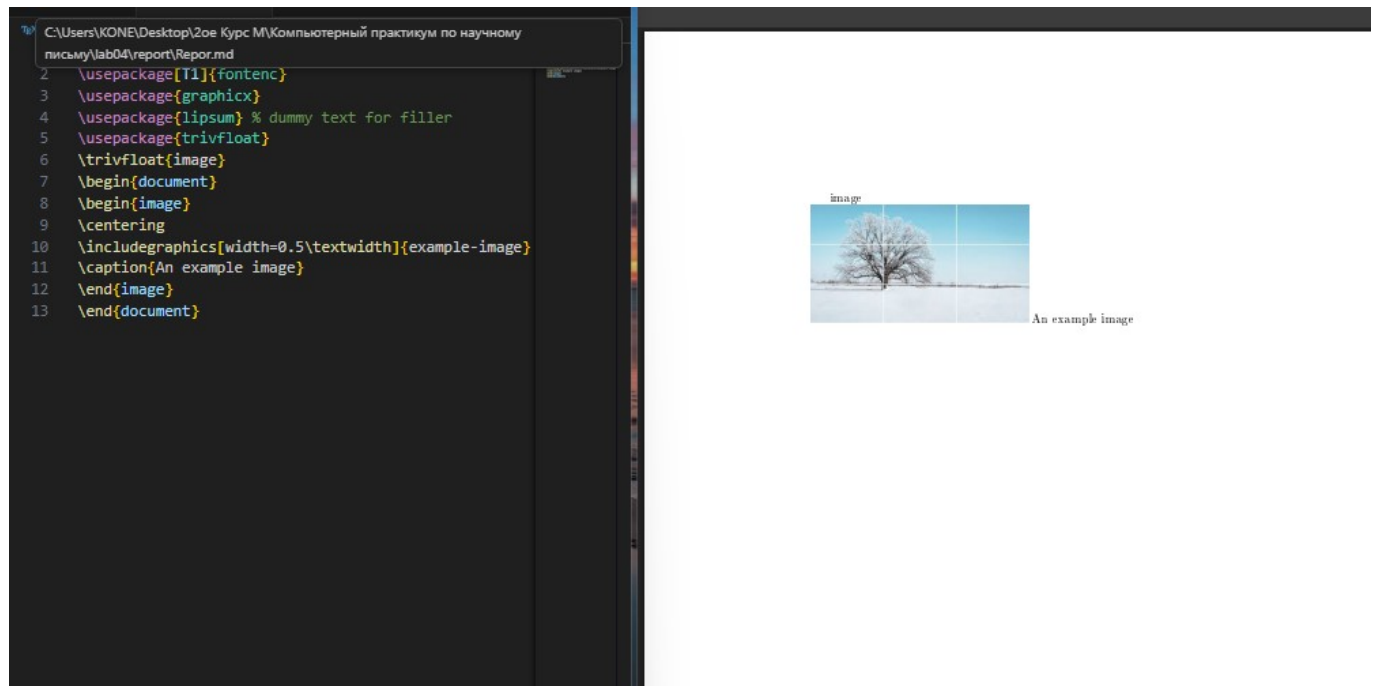


Figure 1: An example image

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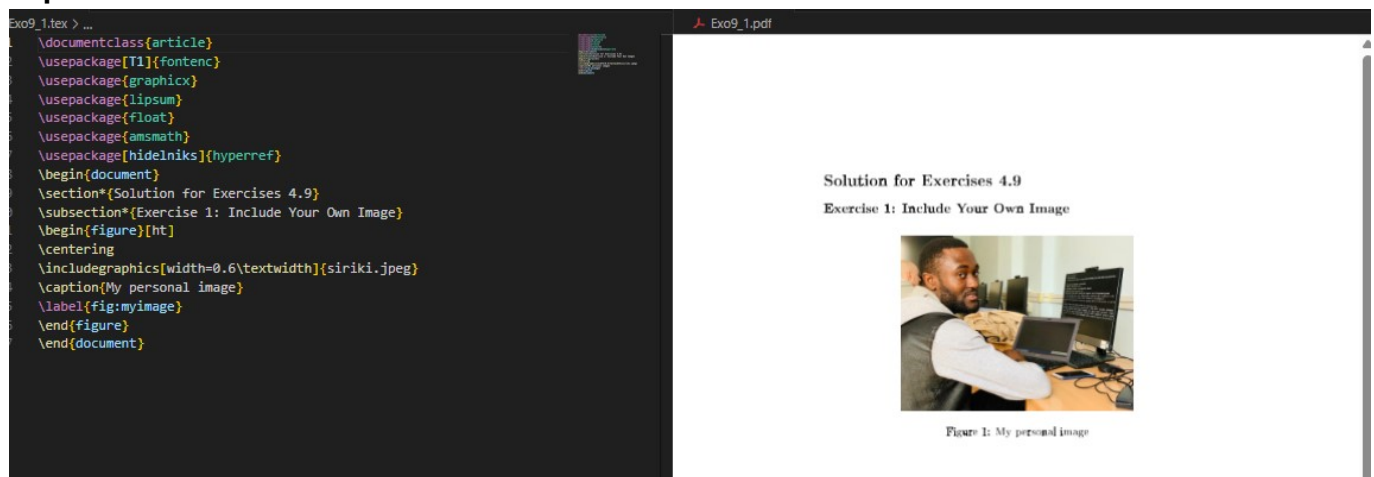


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Выполнение лабораторной работы

Упражнение 1-2: Включение изображений и опции размера

Упражнение 1



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Упражнение 2

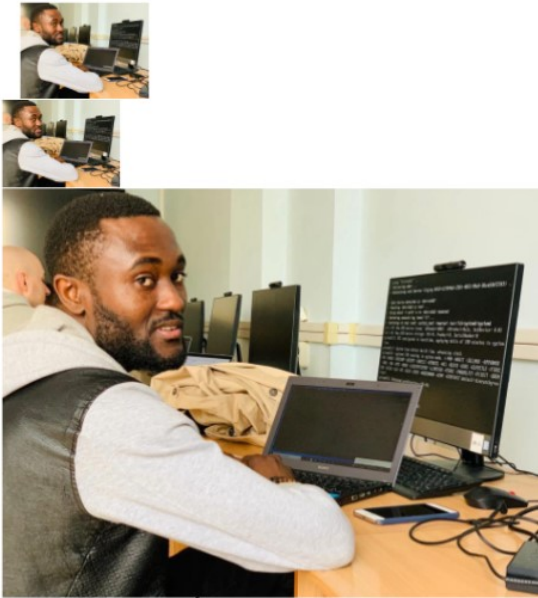
Exo9_2.tex > ...

C:\Users\KONE\Desktop\2oe Курс М\Компьютерный практикум по научному письму\lab04\Exo9_2.tex

```
1 \usepackage{graphicx}
2 \usepackage{lipsum}
3 \usepackage{float}
4 \usepackage{amsmath}
5 \usepackage{hidelniks}{hyperref}
6 \subsection*{Exercise 2: Explore Size and Rotation Options}
7
8 \begin{document}
9
10 % Different formatting options
11 \includegraphics[height=3cm]{siriki} \\
12 \includegraphics[width=0.3\textwidth]{siriki} \\
13 \includegraphics[scale=0.5]{siriki} \\
14 \includegraphics[angle=45, width=0.2\textwidth]{siriki}
15
16 \end{document}
```

Exo9_2.pdf

Exercise 2: Explore Size and Rotation Options



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Упражнение 3-4: Сравнение размеров и размещение

Упражнение 3

Exo9_3.tex (preview @)

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```
1 \usepackage[T1]{fontenc}
2 \usepackage{graphicx}
3 \usepackage{lipsum}
4 \usepackage{float}
5 \usepackage{amsmath}
6 \usepackage{hidelniks}{hyperref}
7 \subsection*{Exercise 3: Compare \textwidth vs \linewidth}
8
9 \begin{document}
10
11 \lipsum[1]
12
13 \begin{figure}[ht]
14 \centering
15 \includegraphics[width=0.8\textwidth]{siriki}
16 \caption{Using \textbackslash textwidth}
17 \end{figure}
18
19 \begin{figure}[ht]
20 \centering
21 \includegraphics[width=0.8\linewidth]{siriki}
22 \caption{Using \textbackslash linewidth}
23 \end{figure}
24
25 \lipsum[2-5]
26 \end{document}
```

Exo9_3.pdf

Exercise 3: Compare \textwidth vs \linewidth

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, fells. Curabitur dictum gravida mauris. Nam arcu libero, nuncius eget, consectetur id, culpatate a, magna. Donec volutate augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris at leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tunc gravida placerat. Integer sapien est, laucis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan efficitur, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.



Figure 1: Using \textwidth

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Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat et, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec aummy

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Упражнение 4

Exo9_4.tex (preview)

```
1 \usepackage[T1]{fontenc}
2 \usepackage{graphicx}
3 \usepackage{lipsum}
4 \usepackage{float}
5 \usepackage{amsmath}
6 \usepackage[hideIniks]{hyperref}
7 \subsection*{Exercise 4: Float Placement with Different Specifiers}
8 \begin{document}
9 \lipsum[1-2]
10
11 \begin{figure}[h]
12   \centering
13   \includegraphics[width=0.4\textwidth]{example-image-a}
14   \caption{Option h (here)}
15   \label{fig:h}
16 \end{figure}
17
18 \lipsum[3]
19 \begin{figure}[t]
20   \centering
21   \includegraphics[width=0.4\textwidth]{example-image-b}
22   \caption{Option t (top)}
23   \label{fig:t}
24 \end{figure}
25
26 \begin{figure}[b]
27   \centering
28   \includegraphics[width=0.4\textwidth]{example-image-c}
29   \caption{Option b (bottom)}
30   \label{fig:b}
31 \end{figure}
32
33 \lipsum[4-8]
34 \textbf{Observed interactions:}
35 \begin{itemize}
36 \item \texttt{[h]} : Tries to place the float at the exact location
37 \item \texttt{[t]} : Priority for top of page
38 \item \texttt{[b]} : Priority for bottom of page
```

Exo9_4.pdf

Exercise 4: Float Placement with Different Specifiers

Figure 4: Option h (here)

Figure 4: Option t (top)

Figure 4: Option b (bottom)

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Упражнение 5-7: Перекрёстные ссылки

Упражнение 5

Exo9_5.tex (preview)

```
1 \usepackage[T1]{fontenc}
2 \usepackage{graphicx}
3 \usepackage{lipsum}
4 \usepackage{float}
5 \usepackage{amsmath}
6 \usepackage[hideIniks]{hyperref}
7 \subsection*{Exercise 5: Cross-references and Number of Compilations}
8 \begin{document}
9
10 \section{Introduction}
11 \label{sec:intro}
12
13 In section~\ref{sec:intro}, we present...
14
15 \subsection{First subsection}
16 \label{subsec:first}
17
18 As seen in subsection~\ref{subsec:first}...
19
20 \begin{enumerate}
21 \item First point
22 \item Second point\label{item:second}
23 \item Reference to point~\ref{item:second}
24 \end{enumerate}
25
26 \begin{figure}[ht]
27   \centering
28   \includegraphics[width=0.5\textwidth]{siriki}
29   \caption{Test figure}
30   \label{fig:test}
31 \end{figure}
32
33 Figure~\ref{fig:test} shows...
34
35 \end{document}
```

Exo9_5.pdf

Exercise 5: Cross-references and Number of Compilations

1 Introduction

In section 1, we present...

1.1 First subsection

As seen in subsection 1.1...

1. First point

2. Second point

3. Reference to point 2

Figure 1: Test figure

Figure 1 shows...

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Упражнение 6

```
Exo9_6.tex > ...
15 \caption{Figure with label before}
16 \end{figure}
17
18 \begin{figure}[ht]
19 \centering
20 \includegraphics[width=0.4\textwidth]{example-image-b}
21 \caption{Figure with label after}
22 \label{fig:after} % AFTER caption -> CORRECT
23 \end{figure}
24
25 Reference to figure-\ref{fig:before} (incorrect)\
26 Reference to figure-\ref{fig:after} (correct)
27
28 \textbf{Result:}
29 \begin{itemize}
30 \item \texttt{\textbackslash label} before \texttt{\textbackslash caption}
31 \item \texttt{\textbackslash label} after \texttt{\textbackslash caption}
32 \end{itemize}
33
34 \end{document}
35
```

Exercise 6: Placing \label Before/After \caption



Figure 1: Figure with label before



Figure 2: Figure with label after

Reference to figure (incorrect)
Reference to figure 2 (correct)
Result:
• \label before \caption -> incorrect reference (usually to previous section)
• \label after \caption -> correct reference to the figure

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Упражнение 7

```
Exo9_7.tex > ...
1 \documentclass{article}
2 \usepackage[T1]{fontenc}
3 \usepackage{graphicx}
4 \usepackage{lipsum}
5 \usepackage{float}
6 \usepackage{amsmath}
7 \usepackage{hidelniks}[hyperref]
8 \subsection*{Exercise 7: \textbackslash label After \textbackslash end{equation}}
9
10 \begin{document}
11
12 \begin{equation}
13 E = mc^2
14 \end{equation}
15 \label{eq:after} % AFTER end{equation} -> WRONG
16
17 \begin{equation}
18 F = ma
19 \label{eq:inside} % INSIDE equation -> CORRECT
20 \end{equation}
21
22 Reference to equation-\ref{eq:after} (incorrect)\
23 Reference to equation-\ref{eq:inside} (correct)
24
25 \textbf{Result:}
26 \begin{itemize}
27 \item \texttt{\textbackslash label} after \texttt{\textbackslash end{equation}}
28 \item \texttt{\textbackslash label} inside the \texttt{equation} env
29 \end{itemize}
30 \end{document}
```

Exercise 7: \label After \end{equation}

$$E = mc^2 \tag{1}$$

$$F = ma \tag{2}$$

Reference to equation (incorrect)
Reference to equation 2 (correct)
Result:
• \label after \end{equation} -> incorrect reference (usually to previous section)
• \label inside the equation environment -> correct reference to the equation

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Выводы

В ходе лабораторной работы №4 я изучил основы включения и управления графикой в документах LaTeX. Освоил работу с пакетом **graphicx**, научился создавать плавающие объекты, управлять их размещением и создавать перекрёстные ссылки на изображения.

In this lab work #4, I learned the fundamentals of including and manipulating graphics in LaTeX documents. I mastered the **graphicx** package, learned to create float objects, control their placement, and create cross-references to images.

Список литературы

Спасибо за внимание
